

Agniveer Vayu Other than Science (Group Y) - 27 July 2022 - Memory Based Paper

English

Question 1

Directions: A word is given followed by four options. Select the option which explains the meaning of the word and mark the answer.

Calculative

Options:

- A. compute
- B. unwise
- C. thoughtless
- D. indiscreet

Answer: A

Solution:

The correct answer is "Compute"



Key Points

- Let's see the meanings of the given words-
 - **Calculative**-determine (the amount or number of something) mathematically.
 - **Compute**- reckon or calculate (a figure or amount).
 - **Unwise**-(of a person or action) not wise or sensible; foolish.
 - **Thoughtless**-(of a person or their behavior) not showing consideration for the needs of other people.
 - **Indiscreet**-having, showing, or proceeding from too great a readiness to reveal things that should remain private or secret.
 - Hence, according to the meanings of the given words, the correct answer is **Option 1 i.e. Compute.**
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Question 2

Directions: A word is given followed by four options. Select the option which explains the meaning of the word and mark the answer.

Nostalgia

Options:

- A. Feeling illness
- B. Longing for the past
- C. Primitive dance
- D. Simple song

Answer: B

Solution:

The correct answer is '**Longing for the past**'.



Key Points

- The word '**nostalgia**' means '**a sentimental longing or wistful affection for a period in the past**'.
 - **Etymology:** Nostalgia = *nostos* (return home) + *algos* (pain)
 - The given word is a **noun**.
 - **Synonyms:** Homesickness, reminiscence, remembrance, wistfulness, etc.
 - **Example:** Some people feel **nostalgia** for their college days.
- So, the meaning given in **option 2)** is the correct meaning of the given word.

Correct answer: *Longing for the past*

Question 3

Select the correct indirect speech.

Foden said to Haaland, "I have seen the coach today"

Options:

- A. Foden told Haaland that he has seen the coach that day.
- B. Foden told to Haaland that he had seen the coach that day.
- C. Foden said Haaland that he had seen the coach that day.
- D. Foden told Haaland that he had seen the coach that day.

Answer: D

Solution:

The correct answer is "**Foden told Haaland that he had seen the coach that day.**"



Key Points

- Let's look at the basic rules to convert this sentence.
- We will remove **inverted commas**.
- We will use "**that**" before the speech.
- The verb in the Reporting verb is changed into "**tell**" or "**told**" with an object otherwise '**say**' and '**said**' are retained.
- The given sentence is in the **Present perfect tense**.
- There **reported verb** is in the past, so we must change the **Present perfect** into the **past perfect**.
- "**Today**" will change to "**that day**"
- Look at the following examples:
 - Sita said, "I have already cooked". **Direct**
Sita said that she had already cooked. **Indirect**
 - Ram said to Shyam, "I have seen her today". **Direct**
Ram told Shyam that he had seen her that day. **Indirect**
- Hence, the correct answer:** *Foden told Haaland that he had seen the coach that day.*



Additional Information

- Look at the basic rules of converting the tenses

Direct	Indirect
Present indefinite	Past indefinite
Present Continuous	Past Continuous
Present Perfect	Past Perfect
Present Perfect Continuous	Past Perfect continuous
Past indefinite	Past Perfect
Past Continuous	Past Perfect Continuous
Past Perfect	No change in Tense
Past Perfect Continuous	No change in Tense

Question 4

In the following question, out of the four alternatives, choose the one which can be substituted for the given words/sentence.

A person who is devoted to eating and drinking in excess.

Options:

- A. Glutton
- B. Gusto
- C. Plume
- D. Plummet

Answer: A

Solution:

The correct answer is "Glutton"



Key Points

- Let's see the meanings of the given words-
 - **Glutton**: an excessively greedy eater
 - **Gusto**: enjoyment and enthusiasm in doing something
 - **Plume**: a long, soft feather or arrangement of feathers used by a bird for display or worn by a person for ornament
 - **Plummet**: fall or drop straight down at high speed

Thus, the correct answer is **Option 1) i.e. Glutton**

Question 5

Choose the most appropriate option and fill in the blank.

The Verb form of 'Satisfactory' is _____ .

Options:

- A. Satisfy
- B. Satisfying
- C. Satisfied
- D. Satisfaction

Answer: A

Solution:

The correct answer is "**Satisfy**"

Key Points

- "**Satisfactory**" is an **adjective** that means "fulfilling expectations or needs; acceptable, though not outstanding or perfect."
 - Example-The teachers seem to think his work is satisfactory.
- "**Satisfy**" is a **verb** that means "to please someone by giving them what they want or need"
 - Example-They have 31 flavors of ice cream enough to satisfy everyone!
- Hence, the correct answer is Option 1.

Additional Information

- "Satisfying" and "Satisfied" are the verb forms of the verb "Satisfy" which can be used as a present participle and past participle respectively. Hence, we can not say that they are the perfect form of the given adjective.
 - Also, "Satisfying" can be used as a gerund.
-

Question 6

Select the correctly spelled word:

Options:

- A. Bredle
- B. Feasible
- C. Prohibision
- D. Yok

Answer: B

Solution:

The correct answer is "**Feasible**"

Key Points

- Let's see the meanings and the spellings of the words given in the options:-
 - **Bridle**-the leather straps that you put on a horse's head so that you can control it when you are riding it
 - **Feasible**-possible and practical to do easily or conveniently.
 - **Prohibition**-the action of forbidding something, especially by law.
 - **Yak**-a large domesticated wild ox with shaggy hair, humped shoulders, and large horns, used in Tibet as a pack animal and for its milk, meat, and hide.
 - Hence, the correct spelling is "**Feasible**" i.e. **Option 2**.
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Question 7

A sentence has been given with a blank to be filled with an appropriate word. Choose the correct alternative.

The butterfly spent the rest of _____ life unable to fly.

Options:

- A. it
- B. its
- C. their
- D. one's

Answer: B

Solution:

The correct answer is '**its**'.

Key Points

- The given blank needs a pronoun to represent the butterfly.
- We need to use a **possessive pronoun** for a noun without a gender.
- Possessive pronouns (also called "absolute" or "strong" possessive pronouns) are *mine, yours, his, hers, its, ours, yours, and theirs*.

- 'Its' is the correct word for the given blank as it is a possessive pronoun used for nouns without gender.
- So, **option 2** is the correct answer.

Correct sentence- *The butterfly spent the rest of its life unable to fly.*

Question 8

Select the correctly spelt word.

Options:

- A. Hierarchy
- B. Hierarky
- C. Hierarchy
- D. Hirarchy

Answer: C

Solution:

The correct answer is **option 3, i.e., Hierarchy**.



Key Points

- Out of the given options, only option 3 shows the correct spelling of the noun '**Hierarchy**'.
 - **Hierarchy** means a system in which members of an organization or society are ranked according to relative status or authority. (एक ऐसी प्रणाली जिसमें किसी संगठन या समाज के सदस्यों को सापेक्ष स्थिति या अधिकार के अनुसार क्रमबद्ध किया जाता है)
 - **Example:** *The caste system in India was a system of **hierarchy** that preyed on the lower classes.*
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Question 9

Direction: Select the most appropriate option to fill in the blank.

We should abide _____ the laws and regulations of our country.

Options:

A. from

B. with

C. by

D. on

Answer: C

Solution:

The correct answer is 'by.'



Key Points

- The meaning of the phrasal verb "**abide by**" is to accept or obey an agreement, decision, or rule.
 - Phrasal verb "**abide by**" का अर्थ है एक समझौते, निर्णय या नियम को स्वीकार या पालन करना।
 - **Example:** *Competitors must abide by the judge's decision.*
- Therefore, the correct answer is **Option 3**.

Correct sentence: *We should abide by the laws and regulations of our country.*

Question 10

Directions: Choose the word that is opposite in meaning to the given word.

Prudent

Options:

A. Perspicacious

B. Polarize

C. Indiscreet

D. Canny

Answer: C

Solution:

The correct answer is **Indiscreet**.

Key Points

- Let's look at the meanings of the given word and marked option:
 - Prudent**-careful and sensible; marked by sound judgment (सावधान और समझदार; ध्वनि निर्णय द्वारा चिह्नित)
 - Indiscreet**-not careful in saying or doing things that should be kept secret. (ऐसी बातें कहने या करने में सावधानी न बरतें जिन्हें गुप्त रखा जाना चाहिए)
 - Let's look at the meanings of the other given options:
 - Perspicacious**-acutely insightful and wise (अत्यंत अंतर्दृष्टिपूर्ण और बुद्धिमान)
 - Polarize**-to cause to vibrate in a definite pattern (एक निश्चित पैटर्न में कंपन करने के कारण)
 - Canny**-showing self-interest and shrewdness in dealing with others (दूसरों के साथ व्यवहार करने में स्वार्थ और चतुराई दिखाना)
 - Hence from the given meanings, we find that **Prudent** and **Indiscreet** are **antonyms**.
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Question 11

Select the correctly spelt word:

Options:

- A. Begining
- B. Soldier
- C. Guidance
- D. Shinning

Answer: B

Solution:

The correct answer is "**Soldier**"

Key Points

- Let's explore the meaning of the correct word:
- Soldier(noun):** A person who is in an army and wears its uniform, especially someone who fights when there is a war. (एक व्यक्ति जो सेना में है और उसकी वर्दी पहनता है, खासकर वह जो युद्ध होने पर लड़ता है।)

Hence, the correct answer is **Option 2**.

Additional Information

- Let's explore the meaning and the spelling of the other words:
- **Beginning**(noun): The first part of something or the start of something. (किसी चीज का पहला भाग या किसी चीज की शुरुआत।)
- **Shining**(adjective): Sending out or reflecting light. (प्रकाश भेजना या परावर्तित करना।)
- **Guidance**(noun): Help and advice about how to do something or about how to deal with problems connected with your work, education, or personal relationships. (कुछ कैसे करें या अपने काम, शिक्षा, या व्यक्तिगत संबंधों से जुड़ी समस्याओं से निपटने के तरीके के बारे में मदद और सलाह दें।)

Question 12

Select the most appropriate **ANTONYM** of the given word.

REPULSIVE

Options:

- A. obstinate
- B. attractive
- C. assertive
- D. offensive

Answer: B

Solution:

The correct answer is **Attractive**.



Key Points

- Let us see the meaning of **Repulsive**:
 - **Repulsive**:arousing intense distaste or disgust
- From the given options, we can say that the word '**Attractive**' is the opposite in meaning.
- Let us see the meaning of **Attractive**:
 - **Attractive**:pleasing or appealing to the senses
- Hence, the correct answer is **Attractive**.



Additional Information

- Let's see the meaning of other given options:

WORDS	MEANING

Obstinate	sticking to an opinion, purpose, or course of action in spite of reason, arguments, or persuasion
Assertive	having or showing a confident and forceful personality
Offensive	causing someone to feel resentful, upset, or annoyed

Question 13

Direction: Choose the most appropriate answer and fill in the blanks.

The adjective form of 'Decide' is _____.

Options:

- A. Decisive
- B. Decision
- C. Decisively
- D. Decides

Answer: A

Solution:

The correct answer is 'decisive.'



Key Points

- **Adjectives** give us more information about people, places, animals and things.
- Some **adjectives** can be formed from nouns, verbs and even other adjectives by *adding a prefix or a suffix*.
- Like, the suffix **-ful, -able, -ible, -ful, -al, -ial, -ical**:
 - **Example:** *Hopeful, Achievable, Incredible, political*
- The adjective form of '**Decision**' is formed by replacing '**-on**' with '**-ve**'.
- So, the adjective form of '**Decision**' is '**decisive**'.
- Therefore, the correct answer is '**decisive**'.



Hinglish

- **Adjectives** हमें लोगों, स्थानों, जानवरों और चीजों के बारे में अधिक जानकारी देते हैं।

- कुछ **adjectives, nouns, verbs** और यहां तक कि अन्य **adjectives** से उपसर्ग या प्रत्यय जोड़कर भी बन सकते हैं।
 - जैसे, **suffix-ful, -able, -ible, -ful, -al, -ial, -ical**:
 - **Example:** *Hopeful, Achievable, Incredible, political*
 - '-on' को '-ve' से बदलने से **'Decision'** का विशेषण रूप बनता है।
 - अतः, **'Decision'** का adjective form **'decisive'** है।
-

Question 14

Directions: Choose the correct article from the given options and mark the appropriate option.

Saina Nehwal is _____ badminton player.

Options:

- A. a
- B. an
- C. the
- D. No article

Answer: A

Solution:

The correct answer is 'a.'



Key Points

- In the given sentence, the noun **'badminton'** is **not a specified noun**.
- **'A'** and **'an'** are **indefinite articles**, which means that they refer to, or introduce, an **unspecified noun**.
 - We use **'a'** before a **consonant sound** (here, **badminton**), and we use **'an'** before a **vowel sound**.
- The **definite article 'the'** is used before singular and plural nouns when the **noun is specific or particular**.
- **"No article"** is used when a **plural countable noun is generic or nonspecific** and when a **noncount noun is generic or nonspecific**.

Hence, the correct answer is **option 1**.

Complete Sentence: *Saina Nehwal is **a** badminton player.*



Additional Information

- There are three articles: **a, an, and the**. Articles are **used before nouns or noun equivalents** and are a type of adjective.

RULES	COUNT NOUNS	NON-COUNT NOUNS
Specific identity not known	a, an	(no article)
Specific identity is known	the	the
All things or things in general	(no article)	(no article)

Question 15

Direction: Choose the correct alternative which can be substituted for the given sentence.

Mapping of earth and its formation

Options:

A.

Geography

B.

Lithography

C.

Calligraphy

D.

Pathology

Answer: A

Solution:

The correct answer is '**Geography**'



Key Points

- From the given option the correct answer is '**Geography**'
- The word is **Geography** is a noun and its meaning is **the study of the features and systems of the earth's surface, including continents, mountains, seas, weather, and plant life, and of the ways in which**

countries and people organize life within an area.

- **Example:** *It's impossible to figure out the geography of this hospital.*
- So this will be the most appropriate answer to substitute the given sentence is **Geography**.

Thus the correct answer is **option 1**.

Additional Information

- Let's understand the meaning of the other given words:

Word	Meaning
Pathology	The scientific study of bodily diseases
Lithography	Printing using a stone or a metal plate with a completely smooth surface
Calligraphy	Art related to ornate, good handwriting



- दिए गए विकल्प से सही उत्तर '**Geography**' है
- शब्द **Geography** एक **noun** है और इसका अर्थ पृथ्वी की सतह की विशेषताओं और प्रणालियों का अध्ययन है, जिसमें महाद्वीप, पहाड़, समुद्र, मौसम और पौधे का जीवन शामिल है, और उन तरीकों से और जिन तरीकों से देश और लोग एक क्षेत्र के भीतर जीवन को व्यवस्थित करते हैं।।
 - **Example:** *It's impossible to figure out the geography of this hospital.*
- तो यह दिए गए वाक्य को बदलने के लिए सबसे उपयुक्त उत्तर होगा **Geography**

इस प्रकार सही उत्तर **option 1** है।

Question 16

Select the word which means the same as the group of words given.

One who leaves one's country to settle in another

Options:

- A. Citizen
- B. Emigrant
- C. Cosmopolitan
- D. Foreigner

Answer: B

Solution:

The correct answer is '**Emigrant**'.



Key Points

- The most appropriate word for the given group of words is '**Emigrant**'.
- It means '**a person who leaves their own country in order to settle permanently in another**'. (उत्प्रवासी; एक व्यक्ति जो स्थायी रूप से दूसरे में बसने के लिए अपना देश छोड़ देता है)
 - Example: *The would-be **emigrant** faced many difficulties.*

Correct Answer: *Emigrant.*



Additional Information

- Let's look at the **meaning** of other words:
 - **Citizen**: a person who is legally accepted as a member of a particular country. (किसी देश का विधिसम्मत निवासी, नागरिक)
 - **Cosmopolitan**: containing people from all over the world (दुनिया भर के लोगों से युक्त)
 - **Foreigner**: a person who belongs to a country that is not your own. (आपने देश से भिन्न देश का निवासी; विदेशी व्यक्ति)

Question 17

Select the correct one word for the given group of words:

One who loves books

Options:

- A. Plagiarist
- B. Bibliophile
- C. Polyglot
- D. Bibliophobe

Answer: B

Solution:

The correct word for this is **Bibliophile**.

Key Points

- Let's have a look at the **meaning and example of the marked option**:
 - **Bibliophile**(noun): "a person who loves or collects books". (एक व्यक्ति जो पुस्तकों से प्यार करता है या इकट्ठा करता है)
 - **Example**: *Amongst many things, she was also known as a bibliophile.*
 - Let's have a look at the **meaning of the other given options**:
 - **Plagiarist**: One who plagiarizes; or purloins the words, writings, or ideas of another, and passes them off as his or her own. Example: *The reporter was shown to be a liar and a plagiarist.*
 - **Polyglot**: one who knows or uses many languages.
 - **Bibliophobe**: One who who fears and hates books, book-learning or reading.
 - Hence, we find that **Bibliophile** is the **correct one-word substitute** for the given meanings.
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Question 18

Change the given sentence into indirect speech.

I say to Candice, "Paris is the capital of France, while Dublin is the capital of Ireland."

Options:

- A. I tell Candice that Paris is the capital of France, while Dublin had been the capital of Ireland.
- B. I tell Candice that Paris is the capital of France, while Dublin is the capital of Ireland.
- C. I tell Candice that Paris was the capital of France, while Dublin is the capital of Ireland.
- D. I tell Candice that Paris is the capital of France, while Dublin was the capital of Ireland.

Answer: B

Solution:

The correct answer is "I tell Candice that Paris is the capital of France, while Dublin is the capital of Ireland."

Key Points

- Let's look at the following rule:

- If the Reporting verb is in the Present or Future or the Reported speech is a universal truth, the tense of the Reported speech does not change.
- **Say to** will change to **tell**.
 - when we use **tell/told** we normally say who is spoken to, so we have to use it with a direct personal object. When we use **say/said**, we don't normally indicate who is spoken to, or, if we do, we must put to before the object
- Look at the following example:
 - **Direct:** I say to her, "I shall give her a car."
 - **Indirect:** I tell her that I shall give her a car.

Hence, the correct answer: *I tell Candice that Paris is the capital of France, while Dublin is the capital of Ireland.*

Question 19

Directions: Write Synonyms of words given in CAPITAL letters.

WANDERER

Options:

- A. native
- B. beggar
- C. tramper
- D. traveller

Answer: D

Solution:

The correct answer is "**traveller**"

Key Points

- Let's look at the meaning and the correct option.
 - **Traveller:** One who travels or has traveled, to distant places.
 - **Wanderer:** A person who travels aimlessly; a traveller.
- Hence from the given meanings, we find that both words express **similar meanings**.
- Let's look at the meanings of the other given words.
 - **Beggar:** a person, typically a homeless one, who lives by asking for money or food.
 - **Tramper:** A person who walks for long distances in rough country for recreation.

- **Native:** a person born in a specified place or associated with a place by birth, whether subsequently resident there or not.
-

Question 20

Direction: Choose the word that is the same as the meaning of the given word.

Confiscate

Options:

A.

Release

B.

Seize

C.

Produce

D.

Exhibit

Answer: B

Solution:

The correct answer is 'Seize'.

Key Points

- The word **Confiscate** means **Take temporary possession of as a security, by legal authority.**
 - **Ex:** "The police **confiscated** the stolen artwork"
- **Let's explore the given options:**
 - **Seize:** Take hold of, grab.
 - **Exhibit:** Show an attribute, property, knowledge, or skill.
 - **Produce:** Bring forth or yield.
- It is clear that **Confiscate** and **Seize** are synonyms of each other.
- Hence, **option 2** is the correct answer.



- इस शब्द का अर्थ है कानूनी अधिकार द्वारा सुरक्षा के रूप में अस्थायी कब्ज़ा करना।
 - **Ex:** "The police **confiscated** the stolen artwork"
 - आइए दिए गए विकल्पों को देखें:
 - **Seize:** पकड़ लेना, पकड़ लेना।
 - **Exhibit:** एक विशेषता, संपत्ति, ज्ञान, या कौशल दिखाएं।
 - **Produce:** आगे लाओ या उपज।
 - यह स्पष्ट है कि **Confiscate** और **Seize** एक दूसरे के पर्याय हैं।
 - इसलिए, **विकल्प 2** सही उत्तर है।
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RAGA

Question 21

If $x + (1/x) = 2$, then what is the value of $x^{64} + x^{121}$?

Options:

- A. 0
- B. 1
- C. 2
- D. -2

Answer: C

Solution:

Formula used:

$$(a - b)^2 = a^2 - 2ab + b^2$$

Calculation:

Given that,

$$x + 1/x = 2$$

$$\Rightarrow (x^2+1)/x = 2$$

$$\Rightarrow x^2 - 2x + 1 = 0$$

By using the above formula,

$$\Rightarrow (x - 1)^2 = 0$$

$$\Rightarrow x = 1$$

$$\Rightarrow x^{64} + x^{121} = 1^{64} + 1^{121}$$

$$\therefore x^{64} + x^{121} = 2$$

Question 22

Travelling at $\frac{4}{5}^{\text{th}}$ of his usual speed, a man is 15 minutes late. What is his usual time to cover the same distance?

Options:

A. 15 minutes

B. 1 hours

C. 75 minutes

D. 45 minutes

Answer: B

Solution:

Given:

Traveling at $\frac{4}{5}$ th of his usual speed, a man is 15 minutes late

Formula used:

Speed = Distance/Time

Calculation:

Let the speed & distance travelled are V & D respectively.

$$(T) = D/V \text{ -----(1)}$$

$$\text{Time with } (4/5) \text{ of speed} = \frac{D}{\frac{4V}{5}} = \frac{5D}{4V}$$

According to the question,

$$\Rightarrow \frac{5D}{4V} - \frac{D}{V} = 15 \text{ min}$$

$$\Rightarrow \frac{D}{V} \left(\frac{5}{4} - 1 \right) = 15 \text{ min}$$

$$\Rightarrow T \times \left(\frac{1}{4} \right) = 15 \text{ min } (\because T = D/V)$$

$$\Rightarrow T = 60 \text{ min} = 1 \text{ hr}$$

Thus, the correct answer is 1 hr.

Question 23

In an exam, a student gets 20% marks and fails by 30 marks. A second student scores 32% marks but gets 42 marks more than the minimum passing marks. Find the passing percentage?

Options:

A. 25%

B. 30%

C. 28%

D. 22%

Answer: A

Solution:

Calculation:

Let the total marks of the exam be n and passing marks be m .

$$\Rightarrow \frac{20n}{100} = m - 30 \text{ ----(1)}$$

$$\Rightarrow \frac{32n}{100} = m + 42 \text{ ----(2)}$$

Doing (2) - (1)

$$\Rightarrow \frac{32n}{100} - \frac{20n}{100} = (m + 42) - (m - 30)$$

$$\Rightarrow \frac{12n}{100} = 72$$

$$\Rightarrow n = 600$$

From equation (1)

$$\Rightarrow \frac{20 \times 600}{100} = m - 30$$

$$\Rightarrow m = 150 = \text{passing marks}$$

$$\therefore \text{Passing percentage} = \frac{150}{600} \times 100 = 25\%$$

Question 24

The ratio of the speeds of A and B is 3 : 4. If A takes 15 min more than B to cover the same distance, how much time does B take to cover the distance?

Options:

A. 45 min

B. 60 min

C. 50 min

D. 40 min

Answer: A

Solution:

Given:

Speed of A : Speed of B = 3 : 4

Formula:

Speed = distance/time

Calculation:

Let speed of A and B be 3a kmph and 4a kmph respectively.

Time taken by B to cover d distance be t hours.

Time taken by A to cover same d distance = $t + 15/60 = (t + 1/4)$ hours

Then,

$$\Rightarrow 3a \times (t + 1/4) = 4a \times t$$

$$\Rightarrow 3t + 3/4 = 4t$$

$$\Rightarrow t = 3/4$$

$$\Rightarrow t = 3/4 \times 60 = 45 \text{ min}$$

$\therefore$ Time taken by B is 45 min.



Shortcut Trick

Speed inversely proportional to the Time

Speed ratio A to B = 3 : 4

Time ratio A to B = 4 : 3

According to the question,

$$\Rightarrow 4 - 3 = 1 \text{ unit}$$

$$\Rightarrow 1 \text{ unit} = 15 \text{ min}$$

So, the time is taken B cover the same distance = 3×15

$$\Rightarrow 45 \text{ min}$$

$\therefore$ Time taken by B is 45 min.

Question 25

What is the probability of getting sum equal to either 3 or 5 while rolling two dices simultaneously?

Options:

A. $1/2$

B. $1/3$

C. $1/4$

D. $1/6$

Answer: D

Solution:

Given:

2 dice are rolled simultaneously.

Formula used:

Probability of occurrence of the event:

$$P(E) = \frac{n(E)}{n(S)}$$

Where,

$n(E)$ = Number of favorable outcome

$n(S)$ = Number of possible outcome

Calculation:

Total number of cases

$$n(S) = 6 \times 6 = 36$$

The possible outcomes are

(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6)

(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6)

(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)

(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)

(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)

(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)

Cases in which the sum is either 3 or 5

(1, 2), (2, 1), (2, 3), (3, 2), (1, 4), (4, 1)

$$\Rightarrow n(E) = 6$$

$$\text{Required probability } P(E) = \frac{n(E)}{n(S)}$$

$$\therefore P(E) = \frac{6}{36} = \frac{1}{6}$$

Question 26

In 1 kg. mixture of sand and iron, 20% is iron. How much sand should be added so that the proportion of iron becomes 10% ?

Options:

- A. 2 kg
- B. 1.5 kg
- C. 1 kg
- D. 0.5 kg

Answer: C

Solution:

Given:

The percentage of iron in 1 kg of mixture is 20%

Calculation:

The amount of iron = $20/100 \times 1000 = 200\text{gm}$ (1kg = 1000 gm)

The amount of sand = $1000 - 200 = 800\text{ gm}$

Let the add amount of sand be x

According to the question;

$$\Rightarrow 200/(1000 + x) = 10/100$$

$$\Rightarrow 2000 = 1000 + x$$

$$\Rightarrow x = 1000\text{ gm} = 1\text{ kg}$$

$\therefore$ The added amount of sand is 1 kg.

Question 27

The selling price of an article is Rs. 356 at 11% loss. What should be its selling price if a shopkeeper wants to earn a profit of 11%.

Options:

A. Rs. 440

B. Rs. 444

C. Rs. 448

D. Rs. 454

Answer: B

Solution:

 Shortcut Trick

$$89\% \text{ of CP} = 356$$

$$\Rightarrow 111\% \text{ of CP} = (111 \times 356)/89$$

$$\Rightarrow 111\% \text{ of CP} = \text{Rs. } 444$$

$\therefore$ In order to gain 11% profit, the selling price is Rs. 444.

 Alternate Method

Formula used:

$$\text{SP} = \{(100 + P\%)/100\} \times \text{CP}$$

$$\text{SP} = \{(100 - L\%)/100\} \times \text{CP}$$

Calculations:

According to the question, we have

The cost price of an article is

$$\text{SP} = \{(100 + L\%)/100\} \times \text{CP}$$

$$\Rightarrow 356 = \{(100 - 11)/100\} \times \text{CP}$$

$$\Rightarrow \text{CP} = (356 \times 100)/89$$

$$\Rightarrow \text{CP} = \text{Rs. } 400$$

In order to 11%, The selling price of an article is

$$\text{SP} = \{(100 + P\%)/100\} \times \text{CP}$$

$$\Rightarrow \text{SP} = \{(100 + 11)/100\} \times 400$$

$$\Rightarrow \text{SP} = (111 \times 400)/100$$

$$\Rightarrow SP = \text{Rs. } 444$$

$\therefore$ In order to gain 11% profit, the selling price is Rs. 444.

Question 28

Two numbers are in the ratio of 15 : 11. If their HCF is 13, then the greater number is:

Options:

A. 195

B. 125

C. 175

D. 120

Answer: A

Solution:

Given:

Two numbers are in the ratio of 15 : 11. If their HCF is 13

Calculation:

Let the number be $15x$ and $11x$

HCF is 13

HCF of two number $x = 13$

Then numbers

$$\Rightarrow 15 \times 13 = 195$$

$$\Rightarrow 11 \times 13 = 143$$

$\therefore$ The greater number is 195.

Question 29

Who among the following won the US Open 2021 women's single title?

Options:

- A. Marina Stakusic
- B. Rebecca Marino
- C. Emma Raducanu
- D. Leylah Fernandez

Answer: C

Solution:

The correct answer is Emma Raducanu.



Key Points

- **Emma Raducanu defeated Leylah Fernandez** to win the US Open women's single (tennis) title in September 2021.
- She has been voted **BBC Sports Personality of the Year 2021**.
- She has become the third WTA (Women's Tennis Association) player to win the award in its 67-year history.
- **Raducanu became the first British woman to win a Grand Slam singles title after defeating Leylah Fernandez** in the Flushing Meadows final in September 2021.



Additional Information

- **Marina Stakusic** is a Canadian tennis player.
 - **Rebecca Catherine Marino** is a Canadian professional tennis player.
 - **Mia Kupres** is a tennis player from Canada.
-

Question 30

What is the full form of USB that we associate with a USB drive ?

Options:

- A. Universal Serial Bus

- B. Unique Service Bus
- C. Uniform Service Bus
- D. Uniform Serial Bus

Answer: A

Solution:

The correct answer is Universal Serial Bus.



Key Points

- A Universal Serial Bus(USB) is the common computer port that allows **communication between computers and other peripheral devices** like mouse, keyboards, scanners, printers, etc.
- The **first version of USB** was introduced in **1996**.
- The USB can be used for various purposes, such as
 1. **To transfer data,**
 2. **To connect peripheral devices,**
 3. **Even as an interface for charging devices such as smartphones, digital cameras, etc.**
- The USB has replaced interfaces like the parallel and serial port because of its versatile nature.



Additional Information

- A USB is intended to allow hot-swapping and enhance plug-and-play between computer and peripheral devices.
- The plug-and-play means that operating systems involuntarily discover and configure the device.
- The hot-wapping allows the replacement and removal of peripheral devices.

Question 31

_____ was the author of the book title "**Wings of Fire**".

Options:

- A. Dr A.P.J Abdul Kalam
- B. Mulk Raj Anand
- C. Jawahar Lal Nehru
- D. Maithili Sharan Gupt

Answer: A

Solution:

The correct answer is A.P.J Abdul Kalam.

- The book 'Wings of Fire' is authored by A.P.J Abdul Kalam.

Key Points

- The book was written by Dr. Abdul Kalam and Arun Tiwari.
- His full name is Avul Pakir Jainulabdeen Abdul Kalam.
- He served as the **11th President of India** from **2002 to 2007**.
- He also helped to establish the **Vikram Sarabhai Space Centre** and pioneered the **first space launch vehicle program**.
- He is known as the **Missile Man of India**.

Additional Information

Name	Famous Creation
Mulk Raj Anand	Coolie, Untouchable
Jawahar Lal Nehru	The Discovery of India
Maithili Sharan Gupt	Panchvati, Urmila

Question 32

In which country, the Pegasus spyware has been developed?

Options:

- A. Israel
- B. Brazil
- C. Russia
- D. China

Answer: A

Solution:

The correct answer is Israel.

Key Points

Pegasus:

- It is a type of malicious software or **malware classified as spyware**.
 - It is designed to gain access to devices, without the knowledge of users, and gather personal information and relay it back to whoever it is that is using the software to spy.
 - Pegasus has been developed by the **Israeli firm NSO Group** that was set up in 2010.
-

Question 33

Who was the first Indian to become Governor General of independent India?

Options:

- A. Rajendra Prasad
- B. Jawaharlal Nehru
- C. Rajagopalachari
- D. Vallabhbhai Patel

Answer: C

Solution:

The correct answer is **Rajagopalachari**.

Key Points

- **C Rajagopalachari** was an Indian politician, freedom fighter and member of the Constituent Assembly from Madras.
- He was a senior leader of the **Indian National Congress**.
- He was **the first and last Indian Governor-General of free India**.
- C. Rajagopalachari **succeeded Lord Mountbatten as governor-general of India from 1948 to 1950**.
- **After C. Rajagopalachari, the title of the governor was abolished and was replaced by the President of India**.
- He was known for the formulation of the proposal to solve the political deadlock between the All India Muslim League and the Indian National Congress in 1944, popularly known as **Rajagopalachari's formula**.
- He was one of the first recipients of the **Bharat Ratna** along with Sarvepalli Radhakrishnan and C. V. Raman, **in 1954**.
- He was the recipient of the **Sahitya Akademi Award** for the Tamil language for his work retelling of the **Ramayana – Chakravarti Thirumagan**.

- He founded the **Swatantra Party** in **1959**.

Additional Information

- **Rajendra Prasad:**
 - **Rajendra Prasad** was the **longest-serving President of India (1950 to 1962)**.
 - **Rajendra Prasad** was elected as the **first President of India on 24th January 1950**
 - He was one of the **prominent leaders of the Indian independence movement** and played a prominent role as the President of the Indian National Congress.
 - During his tenure, **H.C. Mookerjee was the Vice-President**.
 - He was a **lawyer by profession**, and also an influential leader during the Indian freedom struggle.
 - He served as the **president of the Constituent Assembly** that drafted the first constitution of the Republic of India, which lasted from **1948 to 1950**.
 - He also became the **first Minister of Food and Agriculture in 1946** in the **Interim National Government**.
- **Jawaharlal Nehru:**
 - **Jawaharlal Nehru** was the **first Prime Minister of India**.
 - He is the Prime Minister who has had the **longest term in office**.
 - He served as the prime minister of India from **15th August 1947 to 1964**.
 - He was the Prime Minister of India during the **Indo-China war in 1962**.
 - He served as the first chairman of the planning commission of India.
 - He was honoured with the **Bharat Ratna in 1955**.
 - The name **Panchayati raj** was given by Jawaharlal Nehru.
 - **National Herald** is the newspaper started by Jawaharlal Nehru.
 - **Shantivanis** is the cremation ground of Jawaharlal Nehru.
- **Vallabhbhai Patel:**
 - He served as the **First Deputy Prime Minister of India**.
 - He was an Indian barrister and a senior leader of the Indian National Congress who played a leading role in the **country's struggle for independence and guided its integration into a united, independent nation**.
 - He was one of the **conservative members** of the Indian National Congress.
 - In India and elsewhere, he was often called **Sardar**, meaning "**chief**" in Hindi, Urdu, and Persian.
 - He acted as Home Minister during the **political integration of India and the Indo-Pakistani War of 1947**.

Question 34

What is the capital of Ukraine?

Options:

- A. Chişinău
- B. Bucharest
- C. Minsk
- D. Kyiv

Answer: D

Solution:

The correct answer is Kyiv.



Key Points

- **Kyiv** is the capital of Ukraine.
- **Currency:** hryvnia
- **President:** Volodymyr Zelenskyy.



Additional Information

- **Bucharest** is the capital of **Romania**.
 - Currency: **Romanian leu**.
 - **Minsk** is the capital of **Belarus**.
 - Currency: **Belarusian ruble**.
 - **Republic of Moldova:**
 - Its capital is **Chişinău**.
 - Currency: **Moldovan leu**
-

Question 35

Which of the following is the Light Combat Aircraft of India?

Options:

A. Chetak

B. Tejas

C. Rudra

D. Dhruv

Answer: B

Solution:

The correct answer is Tejas.



Important Points

- **Tejas** is the name of an Indian Light Combat Aircraft.

- **Tejas-Indian Light Combat Aircraft (LCA)**, is the smallest and lightest **Multi-Role Supersonic Fighter Aircraft** of its class.
- This single-engine, Compound-Delta-Wing, Tailless Aircraft Tejas is designed and developed by ADA with HAL as the principal partner along with **DRDO, CSIR, BEL, DGAQA**, and IAF to meet the diverse needs of the Indian Air Force (IAF).
- The light combat aircraft was named 'Tejas', which means 'radiance', by former prime minister Atal Bihari Vajpayee.
- Tejas incorporates a distinctive 'glass cockpit' in which information is displayed real-time to the pilot.
- The current **Air Chief Marshal** of IAF is **VR CHAUDHARI**.

Additional Information

- **Sukhoi Su-30** is a two-seater, twin-engine fighter aircraft developed by Russia's Sukhoi Aviation Corporation in the Soviet Union.
 - The Indian Air Force has nearly 360 SU 30 fighter aircraft.
 - **Rudra** is the Weaponised version of the **Advanced Light Helicopter** (Dhruv) designed and developed by **Hindustan Aeronautics Limited (HAL)** to meet the requirements of the Indian Army & Air Force.
 - The indigenously designed and developed Advanced Light Helicopter (**ALH-DHRUV**) is a twin-engine, multi-role, multi-mission new generation helicopter in the 5.5-ton weight class.
-

Question 36

If mirror image shows 4:20 in a 12 hrs clock, then what will be the actual timing?

Options:

- A. 7:30
- B. 7:44
- C. 7:40
- D. 7:45

Answer: C

Solution:

For every mirror image based on a clock,

Given :

The clock shows the time = 4:20

To find :

Time is shown in the mirror image

Concept used :

For mirror image

(11- hours) : (60 - minutes)

Solution :

Hours = 4

Minutes = 20

Mirror image = (11- 4) : (60 - 20)

⇒ Mirror image = 7:40



Additional Information

If it is 12 hrs clock, then subtract the given timing from 11:60 hrs.

If it is 24 hrs clock, then subtract the given timing from 23:60 hrs.

Hence, the real time will be 7:40.

Question 37

Choose the Venn diagram that best represents the three classes given in the question below:

Copper, Book, Wire



(A)



(B)



(C)



(D)

Options:

A. A

B. B

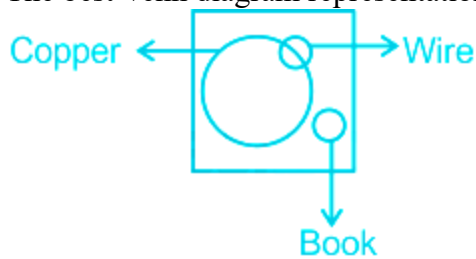
C. C

D. D

Answer: C

Solution:

- Some wires are made of copper.
- A book is not related to copper or wire.
- The best Venn diagram representation is as follows:



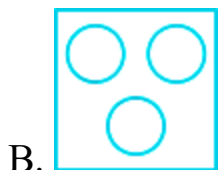
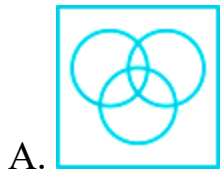
Hence, "option 3" is the correct answer.

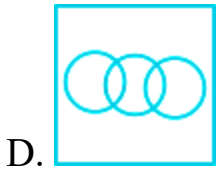
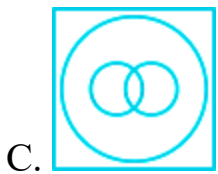
Question 38

Select the option that correctly represents the relationship among the following:

Smokers, Lawyers, Non-smokers

Options:





Answer: D

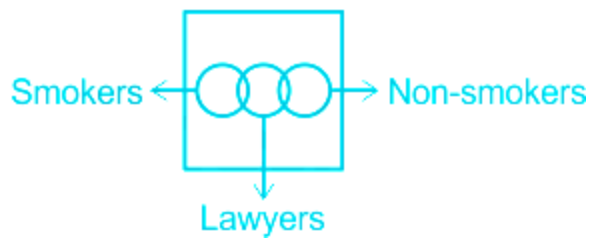
Solution:

Some smokers may be lawyers and vice versa.

Some non-smokers may be lawyers and vice versa.

But smokers are entirely different from non-smokers.

The correct Venn diagram representation is,



Hence, option 4 is the correct answer.

Question 39

Select the number from among the given options that can replace the question mark (?) in the following series.

65, 70, 80, 80, 110, 100, ?

Options:

A. 161

B. 145

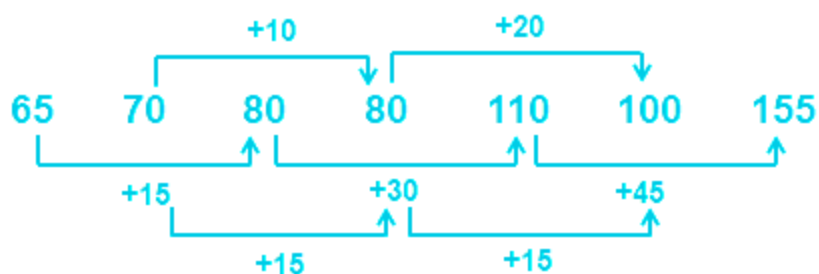
C. 130

D. 155

Answer: D

Solution:

The pattern followed here is:



Hence, '155' is the correct answer.

Question 40

Select a figure amongst the four alternatives that when placed in the blank space(?) of figure X will complete the pattern. (Rotation is not allowed).



Options:





Answer: D

Solution:

The figure which will complete the pattern is:



Hence, option 4 is the correct answer.

Question 41

Shreya starts walking towards the north from her shop. She walks 15 m, then takes a right turn and walks 30 m. She now takes a left turn and walks 15 m, then again she takes a left turn and walks 15 m. Finally, she turns left again and stops after walking 15 m. In which direction is the end point with respect to the starting point? (all turns are 90° turns only)

Options:

A. North-west

B. South-east

C. South-west

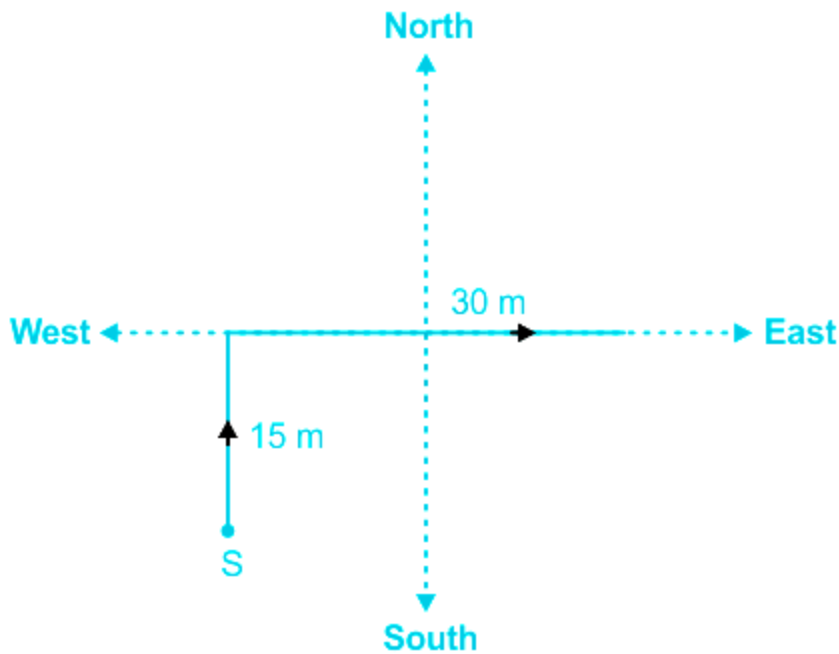
D. North-east

Answer: D

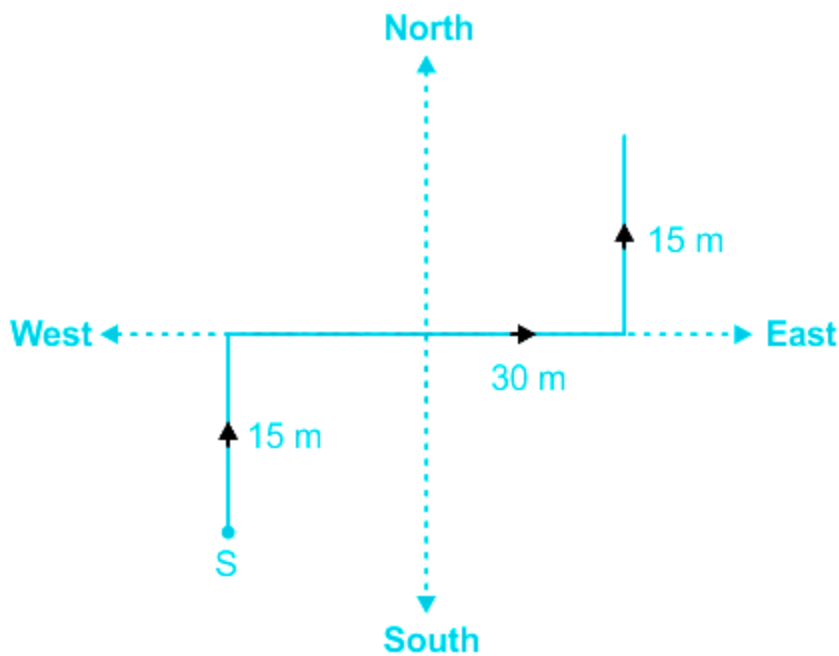
Solution:

Given,

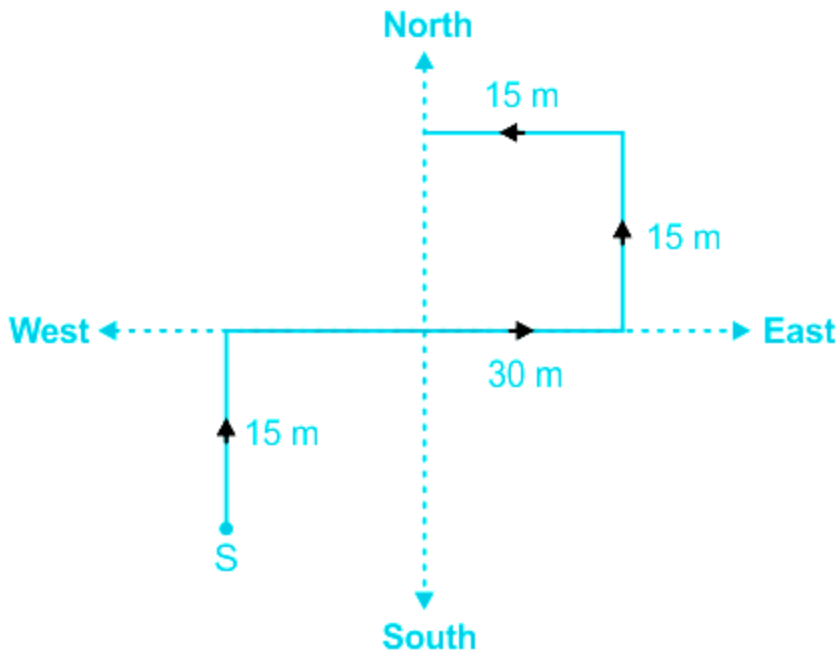
1. Shreya starts walking towards the north from her shop and she walks 15 m. Then takes a right turn and walks 30 m.



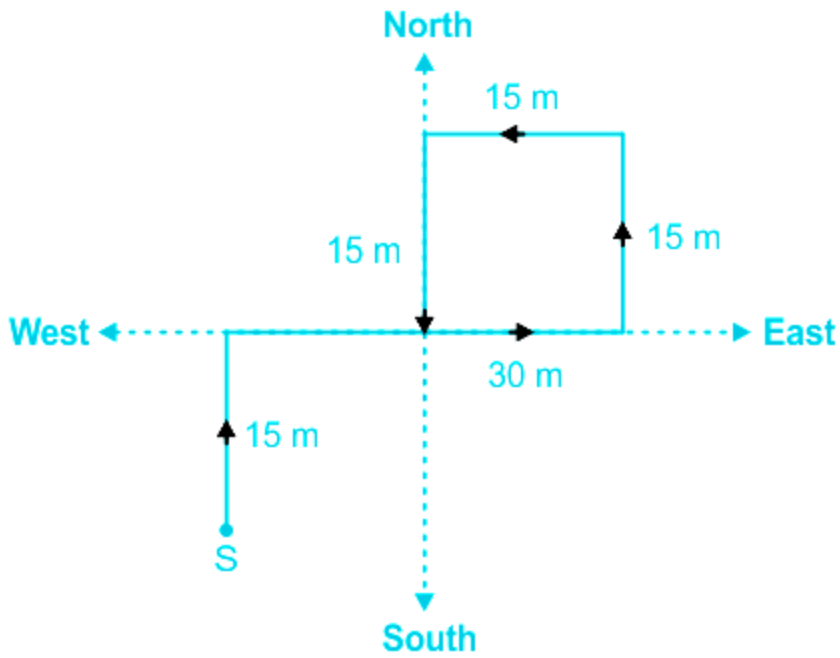
2. She now takes a left turn and walks 15 m.



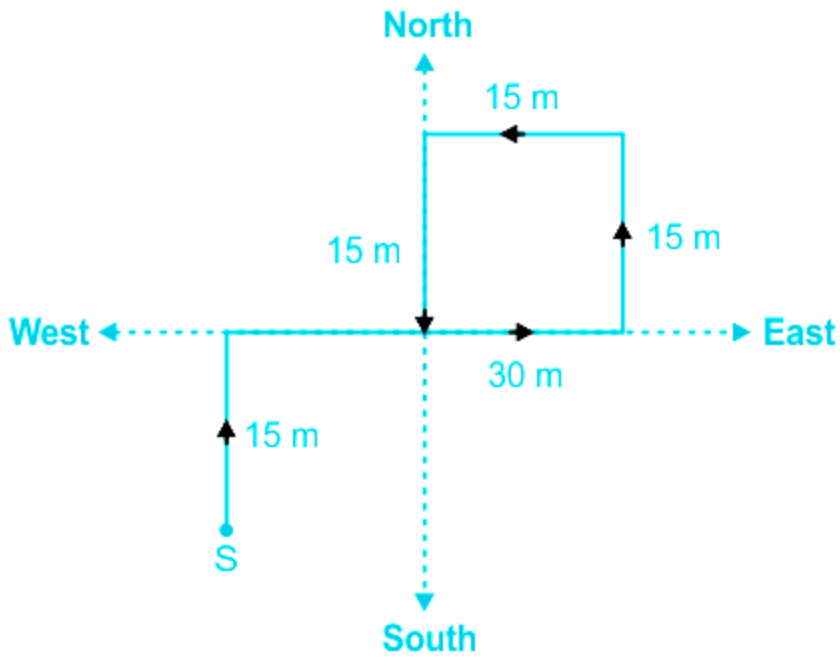
3. Again she takes a left turn and walks 15 m.



4.Finally, she turns left again and stops after walking 15 m.



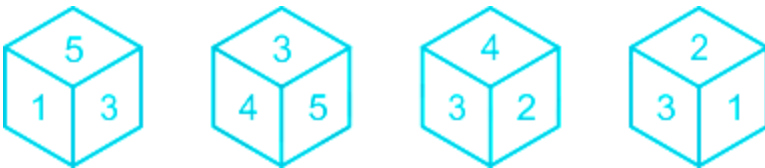
Final Direction according to the given information.



She is in **North-east** direction.

Question 42

Study the different positions of a dice given below. Which numbers will occur on the face opposite to 4?



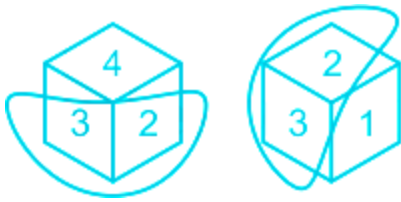
Options:

- A. 3
- B. 5
- C. 1
- D. 6

Answer: C

Solution:

Taking Dice position (3) and (4), as 2 and 3 are common;



The remaining number in Dice position (3) is 4.

The remaining number in Dice position (4) is 1.

So, 1 and 4 are opposite number pairs.

Hence, "1" is the correct answer.

Question 43

Snigda said that she met her mother's only brother's son's mother's father-in-law. Whom did she meet?

Options:

- A. Uncle
- B. Grandfather
- C. Father
- D. Brother

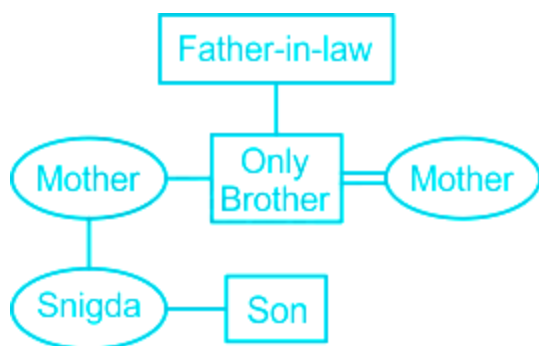
Answer: B

Solution:

From the given information,

Symbol in Diagram	Meaning
○	Female
□	Male
==	Married Couple
—	Siblings
	Difference of A Generation

Snigdha met her mother's only brother's son's mother's father-in-law, implies she met her maternal grandfather as shown below, the final family tree is as follows:



Hence, **grandfather** is the correct answer.

Question 44

In a group of five girls, Komal is the shortest while Lisa is the tallest. Rohini is taller than Komal but shorter than Veena. Priya is the second shortest person girl in the group. Who is the second tallest girl in the group?

Options:

- A. Veena
- B. Lisa
- C. Rohini

D. Priya

Answer: A

Solution:

Given: Five Girls - Komal, Lisa, Rohini, Veena, and Priya

1. Komal is the shortest while Lisa is the tallest.

$Lisa > _ > _ > _ > Komal$

2. Priya is the second shortest person girl in the group

$Lisa > _ > _ > Priya > Komal$

3. Rohini is taller than Komal but shorter than Veena.

$Veena > Rohini > Komal$

By combining all the above statements, we get:

$Lisa > Veena > Rohini > Priya > Komal$

From the above arrangement, Veena is the second tallest girl in the group.

Hence, "**Veena**" is the correct answer.

Question 45

In a certain code language JHMI is written as LPKM. How will YGAN be written in that code language?

Options:

A. QDKA

B. PDJC

C. QDJB

D. PEJB

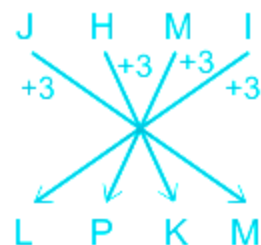
Answer: C

Solution:

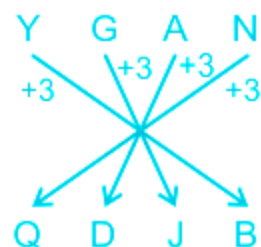
Position values of the letters according to the English alphabet series,

Alphabets	A	B	C	D	E	F	G	H	I	J	K	L	M
Positional value	1	2	3	4	5	6	7	8	9	10	11	12	13
Positional value	26	25	24	23	22	21	20	19	18	17	16	15	14
Alphabets	Z	Y	X	W	V	U	T	S	R	Q	P	O	N

The logic followed here is,



Similarly,



Hence, the correct answer is "QDJB".

Question 46

Select the option that is related to the fifth number in the same way as the second number is related to the first number and the fourth number is related to the third number.

19 : 80 :: 23 : 96 :: 16 : ?

Options:

- A. 76
- B. 68
- C. 70
- D. 62

Answer: B

Solution:

The pattern used in the given pairs :

$$(\text{First term} \times 4) + 4 = \text{Second term}$$

First pair- 19 : 80

$$(\text{First term} \times 4) + 4 = \text{Second term}$$

$$(19 \times 4) + 4 = 80$$

$$(76) + 4 = 80$$

$$80 = 80$$

Second pair- 23 : 96

$$(\text{First term} \times 4) + 4 = \text{Second term}$$

$$(23 \times 4) + 4 = 96$$

$$(92) + 4 = 96$$

$$96 = 96$$

So the third pair becomes,

$$(\text{First term} \times 4) + 4 = \text{Second term}$$

$$(16 \times 4) + 4 = \text{Second term}$$

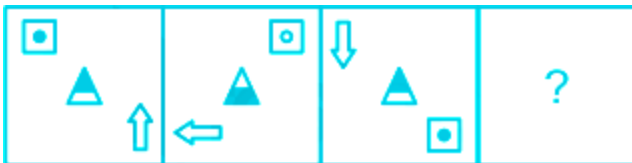
$$(64) + 4 = \text{Second term}$$

$$68 = \text{Second term}$$

Hence the correct answer is 68

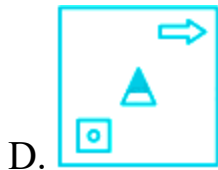
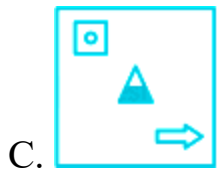
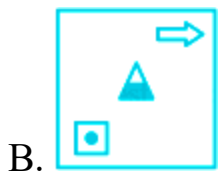
Question 47

Select the figure that will replace the question mark(?) in the following figures series.



Options:





Answer: A

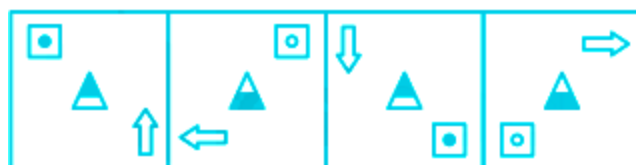
Solution:



The logic followed here is:

-  → upper part shaded in the first step and lower in the next step, thus the same pattern continues.
-  → moves 90° in the clockwise direction, shaded in first and not shaded in next step.
-  → moves 90° in the clockwise direction, and 90° anticlockwise in its own orientation.

The complete series is as follows:



Hence, **option (1)** is the correct answer.

Question 48

Which word cannot be formed from the letters of the word ADMINISTRATION using each letter only once?

Options:

A. SITUATION

B. STRAIN

C. TRADITION

D. RATION

Answer: A

Solution:

ADMINISTRATION → SITUATION cannot be formed from the given word, since there is no U in the word.

ADMINISTRATION → STRAIN can be formed from the given word.

ADMINISTRATION → TRADITION can be formed from the given word.

ADMINISTRATION → RATION can be formed from the given word.

Hence, “situation” cannot be formed using letters of the given word.

Question 49

Select the correct mirror image of the given combination when the mirror is placed at 'PQ' as shown.



Options:

A.

B.

C. $cxYLRgb$

D. $dgЯLRYxo$

Answer: C

Solution:

The correct mirror image of the given combination when the mirror is placed at PQ is as shown below :-



Detailed Explanation:

- The line PQ is the mirror.
- The word "cxYLRgb" the last letter is "b". So, the mirror should be " d " $\Rightarrow$ Option 1 is eliminated.
- The word "cxYLRgb" the second last letter is "g". So, the mirror image should be " g " $\Rightarrow$ Option 4 is eliminated.
- The word "cxYLRgb" the third last letter is "R". So, the mirror image should be " $Я$ " $\Rightarrow$ Option 2 is eliminated.

Hence, the correct answer is "Option 3".

Question 50

Evaluate $8 + 2 \div 3 - 4 \times 6$, if $+$ means $\div$, $\div$ means $-$, $-$ means $\times$ and $\times$ means $+$.

Options:

A. 0

B. 1

C. -2

D. 3

Answer: C

Solution:

According to the BODMAS rule-

B	Brackets in order {}, {}, []	ब्रैकेट {}, {}, [] क्रम में
O	of	का
D	Division (+)	विभाजन (+)
M	Multiplication (*)	गुणा (*)
A	Addition (+)	जोड़ (+)
S	Subtraction (-)	घटाव (-)

Given-

SYMBOL	SIGNS DENOTES
+	÷
÷	-
-	×
×	+

Given equation:-

$$8 + 2 \div 3 - 4 \times 6$$

After replacing the symbols:-

$$\mathbf{8 \div 2} - 3 \times 4 + 6$$

$$= 4 - \mathbf{3 \times 4} + 6$$

$$= \mathbf{4} - 12 + \mathbf{6}$$

$$= \mathbf{10 - 12}$$

$$= - 2$$

Hence, the correct answer is "option (3)".
